

Yukawa Hierarchies from Magnetized Tori without Order-One Coefficients:

Bayesian Selection of the Number of Extra Dimensions

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(Dated: July 9, 2026)

Abstract

Froggatt–Nielsen fits of fermion mass hierarchies involve chosen integer charges and marginalized order-one coefficients. We replace both by geometry: generations are the exact zero modes of a lattice Dirac operator on a magnetized torus (degeneracy = flux = 3), sectors are distinguished by discrete Wilson lines, and Yukawa matrices are computed as overlap integrals with no random coefficients. A single torus is shown to be intrinsically too shallow — its hierarchy depth is set by the flux alone — failing the up-quark hierarchy by two orders of magnitude. The factorizable product $T^2 \times T^2$ squares the suppression and reproduces five of six mass ratios and two of three out-of-sample CKM elements within a factor of five, with Bayesian evidence exceeding the anarchic bound by 15 nats. Treating the geometry itself as a hypothesis space, evidence from mass ratios alone selects three tori over two, while coarse Wilson-line lattices are always preferred; including the CKM elements in the evidence — computed exactly by a triple sum over the shared Wilson line — resolves the apparent mass–mixing tension in favor of three tori on all nine observables. The diagonal generation-pairing ansatz is itself testable: marginalizing over all inter-torus pairings beats it by 3.7 nats even after the Occam penalty, and the pairing proves irreducible to any Wilson-line refinement, mirror, or flux-orientation flip — it aligns Landau towers, not positions. The alternative in which a tilted four-torus Dirac index counts three generations directly, with no pairing at all, is excluded structurally: index 3, being prime, forces one magnetic eigenplane to flatten, capping the attainable hierarchy at the single-tower floor. The pairing is thus established as irreducible discrete data of the compactification, on a par with discrete torsion. All results are reproducible from a dependency-free Rust repository with a machine-validated claim ledger.

I. INTRODUCTION

The fermion mass hierarchy spans five orders of magnitude in the mass ratios and a further structured pattern in the CKM matrix. The Froggatt–Nielsen (FN) mechanism [1] organizes this by integer charges and powers of a small parameter ϵ , but every FN fit carries two kinds of hidden freedom: the *choice* of integer charges, and the *marginalization* over order-one coefficients. A Bayesian model comparison performed within this program (claim QRN-YUK-002) found that a free-charge FN model beats structureless anarchy by $\ln B \geq 23$ even after paying the Occam penalty for $\sim 7.5 \times 10^5$ charge assignments — but the charges

themselves remain a selection, and the $O(1)$ coefficients remain a prior.

This paper asks the sharper question: **can both freedoms be replaced by geometry?** The setting is the standard magnetized-compactification picture [2, 3]: chiral generations arise as degenerate zero modes of the Dirac operator on a torus threaded by Q units of magnetic flux; their number is fixed by the index theorem, their wavefunction profiles by the geometry, and their Yukawa couplings by overlap integrals — with *no random coefficients anywhere*. Our contribution is to make this picture fully computational and fully Bayesian on a lattice, so that:

1. every Yukawa matrix is a deterministic function of a small set of *discrete* geometric data (fluxes, Wilson lines, generation pairing);
2. the geometric hypotheses themselves — how many tori, how fine a Wilson-line lattice, how the generations pair across tori — are placed in the model space and weighed by evidence;
3. negative results (constructions that exist but cannot fit the data) are computed and reported with the same machinery as positive ones.

Point 3 turned out to be the scientific core of the paper. Over the course of the analysis, three “innocent” assumptions were successively promoted to hypotheses and tested: the sorting convention that labels generations (§II D), the diagonal pairing of generations across tori (§V A), and the very idea that pairing is a shadow of the factorized approximation, removable by an honest index-3 geometry (§V B). The first was stabilized after an independent reimplementaion exposed a machine-epsilon knife edge in the published numbers; the second was *rejected by the data* at 3.7 nats; the third was excluded by a structural argument in which the primality of 3 plays the decisive role. What survives is a sharp statement: **the generation pairing is irreducible discrete data of the compactification.**

All numbers in this paper are generated by a dependency-free Rust code base (single shared numerical library, fixed seeds, built-in PASS/FAIL verification against exact results), and every claim is tracked in a machine-validated ledger with explicit evidence files and limitations. The full suite currently reports 314 PASS / 0 FAIL across 59 programs.

II. CONSTRUCTION

A. Zero modes on the lattice

We work on an $N \times N$ lattice torus ($N = 18$ throughout the two-dimensional analysis) with a $U(1)$ hopping Hamiltonian in the Landau gauge, $A_y = \phi x$ with $\phi = 2\pi Q/N^2$. The $Q = 3$ lowest states form an exactly degenerate band (spread $\sim 10^{-13}$) separated by a finite gap (0.115 at $N = 18$) — the lattice avatar of the index theorem’s Q zero modes (claim QRN-MATTER-001). These three states are the three generations.

B. Localization and Wilson lines

Within the degenerate band we diagonalize the position operator $\hat{X} = e^{2\pi i x/N}$; a generic phase ($\phi_0 = 0.83$) avoids the accidental cosine degeneracy of the symmetric choice. The resulting generation wavefunctions are Gaussians of width ≈ 2.9 sites centered 6 sites apart. A discrete Wilson line shifts all centers rigidly — one lattice site per unit — giving each matter sector (Q, u, d, L, e) an *address*: its Wilson-line integer $k \in \mathbb{Z}_6$ (or half-integers for the $\mathbb{Z}_{12}$ refinement).

Development record. Our first implementation attempted lattice magnetic translations to realize sector shifts; these close only when $N \mid 2Q$, which fails at $(N, Q) = (18, 3)$. Wilson lines implement the same physics exactly and are the construction we retain. We report this dead end because the obstruction (a lattice-specific number-theoretic condition) recurs in §V B in a stronger form.

C. Yukawa couplings

The Higgs is a periodic Gaussian of width σ_H centered at the origin; Yukawa matrices are the overlap integrals $Y_{ij} = \sum_x \bar{\psi}_i^{(a)} \psi_j^{(b)} \phi_H$. Given the fluxes, the Wilson lines, and σ_H , *every entry of every Yukawa matrix is determined*. The likelihood on the six mass ratios ($m_u/m_t, m_c/m_t, m_d/m_b, m_s/m_b, m_e/m_\tau, m_\mu/m_\tau$) and, where stated, the three CKM magnitudes ($|V_{us}|, |V_{cb}|, |V_{ub}|$), is lognormal with $\sigma = \ln 2$ — identical to the FN baseline of QRN-YUK-002, so evidence values are directly comparable across the paper.

D. A reproducibility lesson: label stability

The localized modes are sorted by center to align generation labels across sectors. One mode sits *exactly* on the wrap boundary of the coordinate (computed center $\sim 10^{-12}$ from $0 \equiv N$), so the sort is a machine-epsilon knife edge: the published product-model evidences are conditioned on which side of the rounding the implementation happens to fall. An independent numpy reimplement (built to validate the figures) reproduced the single-torus numbers exactly and *disagreed* on the product model by 1.65 nats, which is how the knife edge was found (claim QRN-YUK-006). The fix — snap centers to the half-site lattice before sorting — makes labels convention-stable; the geometry-selection winner and complete ranking are unchanged on both sides of the edge. We adopt the stable convention throughout and quote published-convention values where they are the historical reference. *Fixed seeds do not imply convention-independent reproducibility; degenerate-band sort orders can become physics downstream.*

III. THE SINGLE-TORUS NO-GO

The depth of the single-torus Yukawa hierarchy is set by the flux alone: the attainable minimum of σ_1/σ_3 at $\sigma_H = 1$ is $\approx 3 \times 10^{-3}$ (floor at 2.98×10^{-3} , lattice-size invariant), two orders of magnitude short of $m_u/m_t = 1.3 \times 10^{-5}$. Its full evidence is $\ln Z = -53.8$, *below* the rigorous anarchic upper bound of -35.4 — a structureless model with 18 random coefficients explains the data better than a single magnetized torus with zero random coefficients (claim QRN-YUK-003). This is a principled negative result: geometry with too little structure is worse than no structure.

IV. THE FACTORIZABLE PRODUCT $T^2 \times T^2$

Two tori square the suppression: with per-sector Wilson lines on each factor (6^{10} configurations) and the diagonal generation pairing (each generation is the product of same-label modes), the attainable set reaches the up-quark ratio. The maximum-a-posteriori (MAP) geometry reproduces five of six mass ratios within a factor of five (the exception is m_c/m_t , ratio 14 — a persistent weak spot we report rather than repair) and, *out of sample*, two of three CKM magnitudes ($|V_{cb}|$ within 3%). The evidence is $\ln Z = -20.4$ (published conven-

TABLE I. Model comparison on the six mass ratios.

model	free parameters	$\ln Z$ (6 ratios)
M0 anarchy	18 $O(1)$ coefficients	≤ -35.4 (bound)
M1 FN free charges	10 integers + 18 $O(1)$	-12.2
M2 FN literature charges	18 $O(1)$	-7.6
M2geo single T^2	5 Wilson + σ_H	-53.8
M2geo ² $T^2 \times T^2$	10 Wilson + σ_H	-20.4

tion; -18.8 stable): **15 nats above the anarchic bound with zero random coefficients** (QRN-YUK-003). Against the free-charge FN model ($\ln Z = -12.2$) the residual is -8.2 nats, which we attribute to the coarseness of the Wilson lattice and the pairing ansatz — quantified next.

V. SELECTING THE GEOMETRY ITSELF

Treating $\{\text{number of tori}\} \times \{\text{Wilson lattice}\}$ as a hypothesis space with uniform priors: evidence from the six mass ratios selects T^3 **over** T^2 **by** $+3.0$ **nats**, and every lattice refinement ($\mathbb{Z}_6 \rightarrow \mathbb{Z}_{12}$) *loses* — the data pays for more dimensions but never for finer Wilson lattices (QRN-YUK-004). The T^3 mass-only MAP, however, loses the CKM structure badly, while a 9-observable point evaluation prefers T^2 : an apparent mass-mixing tension.

The tension is an artifact of point estimation. Computing the *evidence* on all nine observables — exact despite the broken factorization, via a triple sum over the shared left-handed Wilson line with the lepton sector factorized — T^3 wins again (-25.56 vs -27.13 published; -21.84 vs -23.61 stable; QRN-YUK-005). The joint MAP places eight of nine observables within a factor of five. Evidence integrates over the attainable set; a bad point does not condemn a good model. This is the third instance in this program (after the anarchic-evidence and calibration episodes) where the *measurement method*, not the model, generated the apparent physics.

A. The generation pairing is a physical degree of freedom

The product construction silently assumed *diagonal pairing*: generation i on torus 1 pairs with generation i on torus 2. Since the $(3, 3)$ -flux T^4 actually has nine zero modes, choosing three diagonal products is a projection ansatz. We promote the pairing to a hypothesis: $\sigma_F \in S_3$ per matter field (gauge-fixing the global relabeling), with uniform prior — an Occam penalty of $4 \ln 6 = 7.17$ nats.

The data rejects the diagonal ansatz. Marginalizing the pairings *gains* 3.7 nats net of the penalty, on masses alone and on all nine observables (QRN-YUK-007). The MAP pairing is non-diagonal. The geometry selection survives: with pairings marginalized on both sides — the T^3 case requiring a certified-truncation evidence sum over $\sim 10^{13}$ terms, with a rigorous remainder bound of width $\sim 10^{-10}$ nats — three tori still beat two (-17.89 vs -19.86 ; QRN-YUK-008).

What *is* the pairing, geometrically? An exhaustive classification against 54 candidate realizations (all $\mathbb{Z}_{18}$ Wilson refinements, mirror reflection, complex conjugation = flux-orientation flip) matches none of the 30 non-trivial pairing states (QRN-YUK-009): the pairing permutes *which Landau towers are bundled into one four-dimensional field*, an internal alignment invisible to position or orientation. A uniform $\mathbb{Z}_3$ shift-orbifold does derive the $9 \rightarrow 3$ projection — the projector’s eigenspaces are exactly anti-diagonal pairing families — but its gauge-invariant prediction (all sectors in the same family) loses to the mixed-parity data by 2.0 nats (QRN-YUK-010); a field-dependent magnetization-orientation model narrows the gap but stops 1.0 nat short of the abstract S_3 pairing at identical prior volume (QRN-YUK-011). The mechanism ladder — diagonal (-23.6) < uniform orbifold (-21.8) < orientation (-20.9) < S_3 pairing (-19.9) — measures in nats how far each physical dressing falls short.

B. The honest-index alternative, and why the prime 3 kills it

One alternative would dissolve the pairing entirely: tilt the four-torus flux so that the Dirac index itself equals three. With flux data (Q_1, Q_2, t, s) on the coordinate 2-planes, the index is $\text{Pf}(F) = Q_1 Q_2 + ts$; $(2, 2, 1, -1)$ gives $4 - 1 = 3$, realized on the lattice as an exactly three-fold degenerate lowest band (spread 10^{-13} , verified against a $t = 0$ control

with degeneracy 4). Its zero modes are non-factorized across the tori (inter-torus mutual information > 0): the geometry itself bundles the towers. If this construction fit the data, σ would be exposed as a shadow of the factorized approximation.

It does not fit, for two instructive reasons.

Lattice index arithmetic. Exact lattice degeneracy is *not* implied by the continuum index: of seven index-3 flux matrices scanned, only three retain an exact triple at $N = 18$, and the flux $(2, 2, 1, -1)$ that is exact at $N = 6$ *splits* at $N = 18$ by 4×10^{-2} . Which (flux, N) pairs protect the degeneracy is a number-theoretic question we leave open (QRN-YUK-015, finding 1).

The prime-3 flattening. The flux two-form decomposes into two magnetic eigenplanes with skew-eigenvalues f_+ , f_- , and the index constraint $f_+ f_- = 3$ with 3 prime forces extreme asymmetry: strengthening the fluxes flattens the second eigenplane ($f_- = 3/f_+ \rightarrow 0$), so suppression only ever acts once. Every exactly-degenerate member of the family sits at the *single-tower* depth floor ($\ln r_1 \in [-5.8, -5.2]$, versus the -11.3 required), and the best evidence achieved — with a purpose-built sparse Lanczos eigensolver scaling the computation to 2×10^5 dimensions at $6,700\times$ the dense-Jacobi speed — is $\ln Z_9 = -34.9$, fifteen nats short of the S_3 -pairing model (QRN-YUK-012–015). The $T^2 \times T^2$ ansatz achieves its depth precisely by *multiplying* two tower suppressions, which an honest index-3 geometry cannot do **because 3 is prime**.

Conclusion of the arc. The pairing survives every attempted reduction — translation, reflection, conjugation, orbifold projection, orientation, tilt — with the last alternative excluded structurally rather than numerically. We conclude that the generation pairing is **irreducible discrete data of the compactification**, of the same standing as discrete torsion or Wilson-line moduli: “why this pairing” is henceforth the same kind of question as “why this flux.”

VI. LIMITATIONS AND OUTLOOK

- m_c/m_t remains the single factor-5 outlier of the MAP throughout (ratio 6–14 depending on the model) — the known weak spot of the M2 class.
- The scan window is $\{T^1, T^2, T^3\} \times \{\mathbb{Z}_6, \mathbb{Z}_{12}\}$, identical tori ($N = 18$, $Q = 3$), a single Gaussian Higgs profile, and CP phases unused. The full-9 evidence excludes $T^3 \times \mathbb{Z}_{12}$

(a 5×10^9 -term triple sum) explicitly.

- The prime-3 argument is a structural statement about magnetic eigenplanes, not a lattice theorem; the number theory of exact lattice degeneracy (which (flux, N) protect the index) is open and, we believe, interesting in its own right.
- The upstream discrete data — fluxes, Wilson lines, and now the pairing — await a common selection principle (moduli stabilization / vacuum selection), which lies beyond the present methods.

Addendum (v17.1). Both deferred questions — the unused CP phases and the selection principle — are taken up in the companion paper [8]: CP violation proves to *require complex structure* (the rectangular geometry’s $J = 0$ is a structural zero, reversed by +306 nats once J is admitted as an observable, with the $|V_{td}|$ holdout miss of this paper’s model turning into a 5% hit), and the first surviving data-blind selection principle reduces, on dissection, to a measure correction on the configuration space.

VII. REPRODUCIBILITY

Everything is deterministic: no random Yukawa coefficients anywhere, fixed seeds where sampling occurs (Bayesian baselines), a dependency-free Rust implementation with a single shared numerical library, and built-in [PASS]/[FAIL] verification in every program (314 PASS / 0 FAIL at the time of writing). Three practices proved load-bearing and we commend them: (i) *claim ledgers* — every statement carries a grade (C0 premise ... C5 interpretation), evidence files, and explicit limitations, machine-validated in CI; (ii) *independent reimplementation* — a figure-generation cross-check in a different language found the label knife edge of §IID; (iii) *certified truncation* — evidence sums too large to enumerate are bounded rigorously, with the remainder interval reported (widths $\sim 10^{-10}$ nats here). Errata are appended, never overwritten: two of our own intermediate claims (a Wilson-refinement equivalence conjecture and an early tolerance choice) were corrected in the record by later

versions.

- [1] C. D. Froggatt and H. B. Nielsen, “Hierarchy of quark masses, Cabibbo angles and CP violation,” Nucl. Phys. B **147** (1979) 277.
- [2] C. Bachas, “A way to break supersymmetry,” arXiv:hep-th/9503030.
- [3] D. Cremades, L. E. Ibáñez and F. Marchesano, “Computing Yukawa couplings from magnetized extra dimensions,” JHEP **05** (2004) 079, arXiv:hep-th/0404229.
- [4] R. Blumenhagen, B. Körs, D. Lüst and S. Stieberger, “Four-dimensional string compactifications with D-branes, orientifolds and fluxes,” Phys. Rept. **445** (2007) 1–193.
- [5] L. E. Ibáñez and A. M. Uranga, *String Theory and Particle Physics: An Introduction to String Phenomenology*, Cambridge University Press (2012).
- [6] R. Trotta, “Bayes in the sky: Bayesian inference and model selection in cosmology,” Contemp. Phys. **49** (2008) 71–104.
- [7] N. Haba and H. Murayama, “Anarchy and hierarchy: An approach to study models of fermion masses and mixings,” Phys. Rev. D **63** (2001) 053010, arXiv:hep-ph/0009174.
- [8] This program, companion paper: “CP Violation Requires Complex Structure,” paper/cp-complex-structure-full.md (LaTeX version in preparation).
- [9] This program’s repository: Quantum Relational Network (`claims.yml`, `results/`, `figures/`) — primary source for all numbers.